

Department CS& IT – MCA (General)

College Management System with Recommender System using Machine Learning

Final Viva Presentation
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Abstract

This project aims to integrate all the essential features of a college, such as conducting online meetings, exams, fee payments, managing notes, and uploading assignments, into a unified platform. The proposed web application will process all the user inputs and utilize predictive analytics to forecast future conditions. By combining various functionalities, this project seeks to enhance the user experience and provide an all-in-one solution for college-related tasks.

Introduction

This project is based on COLLEGE MANAGEMENT SYSTEM. It manages the college information, student information, various different types of events going on in our college. It also keeps track records of all the information regarding students those who are placed in the various organization. It has a notice board which contains information about various cultural or technical or any sports which is supposed to be held soon. Professors can upload notes, conduct online exams, students can upload assignments, parents can submit fees.

This project will also make predictions on students mark, capital loss and gain of next year, trends in courses, scope of a course in future or next year, student's performance measure, teacher's performance measure and placement predictions based on current trends, future scope and Machine Learning algorithms.

Literature Study

- In 2022 Sakshi Naikwadi proposed College Management Web Application System Using Mean Stack. This system provides reliability, time savings, and easy control. Students and their parents can also use this application to view score, attendance, and curriculum details. Students can also view details and notifications anytime, anywhere. This application greatly simplifies and speeds up the process of processing and managing results. This reduces the effort and resources required in traditional processes.
- In 2020 Abhinav Sekhri proposed School Management System which is introduced mainly for School. This system includes functionality like classes, holidays, reports, accounts etc. On the other hand it do not contain library management module from where students as well staff can issue books related to their interest.
- In 2018 Zainab Hussein Arif , Nabeel Salih Ali , Nurul Azma Zakaria , Mohammed Nasser Al-Mhiqani, proposed an attendance management system for educational sectors to keep attendance records, some government and educational institutions in many countries still rely on a paper-based attendance approach. This approach has several drawbacks: waste of time and environmental resources. These traditional attendance tracking methods need to be replaced with more efficient ones. Therefore, much work has been done in this direction. In addition, this survey aims to analyze the latest surveys on automated attendance systems from a scheduling perspective. Our critical review highlighted research in existing literature on technology, application domains, and key findings. It also emphasizes most of the numerous studies on any of the previous three aspects.

Literature Study

- Another similar system, College department Management System (2018) proposed by Ms. A. V. Sinhasane, Ms. A. N. Kashid, Ms. P.J.Kumbhar, Ms. P. R. Shirpale, Prof. S. L. Mortale from International Research Journal of Engineering and Technology which is introduced to reduce the stress and efforts of a staff as well as students. This system have functionalities like voting event details, feedback, newslane etc. This system is basically useful for students as they get the event details through SMS. But there are no basic modules which are important to both staff as well as students such as marks, assignments, notes etc.
- Author Kartiki Datakar (2016), from International Journal of Computer Science and Mobile Computing, quoted that Online Attendance and Feedback System is software developed for daily student attendance in schools, colleges, and institutes. It facilitates to access the information of a particular student in a particular class. It is concluded that a graduated approach to result monitoring is the most effective response, in which sanctions have a place, although only as a last resort. Online Attendance and Feedback System is the software developed for daily student attendance in schools, colleges, and institutes.

Literature Study

- Author 'Lalit Joshi', "A Research Paper on College Management System" (2015), International Journal of Computer Applications, referred that the system utilizes user authentication, displaying only information necessary for an individual's duties. Additionally, each sub-system has authentication allowing authorized users to create or update information in that subsystem. All data is thoroughly reviewed and validated on the server before actual record alteration occurs. In addition to a staff user interface, the system plans for student user interface, allowing users to access information and submit requests online thus reducing processing time.
- In 2012 JinMei-shan, QiuChang-li, LiJing published a book that describes the student information management system using the B / S structure, the system design principle, the system plan and system structure, and the functional modules of the information system according to the current university student information management requirements. It informs large numbers of students and provides an interactive student management platform for managing students.

Literature Study

- In 2010 Zhi -gang YUE, You-wei JIN published a paper discussing the method of the management information in higher education. On the basis of a comprehensive investigation and analysis on the student management in higher education, it established the models of the college students' management information by adopting the advanced information technology, and construct the student management information platform. Moreover, it analyze the characteristics of the information management in higher education, and elaborate the methods to solve the difficulties confronting in the students management of the higher education. Finally, the key method and technology to carry out the information management platform are presented.

Problem Analysis

1. **Student Information Management:** Store and manage student data including personal information, academic history, and enrolment status.
1. **Course Management:** Plan, schedule and manage courses, classes and academic programs.
1. **Finance and Accounting:** Manage college finances, including tuition fee payment, scholarships, and other student financial records.
1. **Attendance and Grade Tracking:** Monitor student attendance and track grades for each student
2. **Staff Management:** Maintain employee records, manage payroll, and monitor employee performance.

Problem Analysis

- 6. **Admissions and Enrolment:** Manage the admissions process and enrolment of new students.
- 6. **Library Management:** Manage library resources, including books, journals, and multimedia materials.
- 6. **Communication and Notification:** Send notifications, alerts and updates to students, staff and other stakeholders.
- 6. **Reporting and Analytics:** Generate reports and analytics on student performance, finance, enrolment, and other critical metrics.

Domain Analysis

The following are the key domains to be analyzed in a CMS :

1. **Student Information Management:** Store and manage student data including personal information, academic history, and enrolment status.
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Functional Requirements

1. Management

Management is the overall controller of the system, they can perform:

1. Change Principal
2. Add HOD
3. Add Professor
4. Add Course
5. Add Lab Assistant
6. Add Librarian
7. Update Fees
8. Edit Personal Details

1. Management

10. Trends in courses

11. Scope of a course in future or next year

2. Users

User contain sub-modules like :

1. Principal
2. HOD
3. Professor
4. Lab Assistant
5. Students
6. Parents

2.1. Principal

Principal controls the entire college activity. Principal can perform:

1. Verify Student
2. View Student
3. View HOD
4. View Professor
5. View Lab Assistant
6. View Librarian
7. Active Meeting Verify Course
8. Edit Personal Details

2.2. HOD

HOD controls the entire department. HOD can perform:

1. Verify Student
2. View Student
3. View HOD
4. View Professor
5. View Lab Assistant
6. View Librarian
7. Active Meeting Verify Course
8. Edit Personal Details

2.2. HOD

9. Student Results prediction

10. Forecast Course Trends

2.3. Professor

Professor manages the class. The Professor can perform:

1. Add and View Students
2. Add Parent
3. Handle the attendance
4. Manage Online Exam
5. Manage Assignment Works
6. Upload Offline Exam, Lab and Assignment Marks Live Meeting
7. Register Lab Complaint
8. Add Notes
9. View Internal Mark

2.3. Professor

10. Edit Personal Details

11. Students mark prediction

2.4. Lab Assistant Lab

Assistant manages the Lab. He/she can perform:

1. Complaint View and Fix
2. Edit Personal Details

2.5. Student

Student can perform:

1. Download Notes
2. Attend Online Exam
3. View Attendance, Online Exam mark, Internal Mark
4. Upload Assignment
5. Pay Semester Fee
6. Live Meeting
7. Edit Personal Details

2.6. Parents

Parents can performs :

1. View Student attendance, Online Exam Mark, Internal Mark
2. Live Meeting
3. Edit Personal Details

3. Librarian

Librarian can manage the entire library. He/She can performs:

1. View Books
2. Add New Books
3. Handle Books
4. Edit Personal Details

4. Office

Office can update the student fee payment. He/She performs:

1. Fee Management
2. Edit Personal Details

5. Placement Cell

Placement cell will manage the placement for the students. He/She performs:

1. Identify job trends
2. Post placement opportunities
3. Manage Companies
4. Edit Personal Details

Non-Functional Requirements

1. Performance

This website uses amp pages for fast loading of webpages and have a lighthouse score of above 70.

2. Security

All logins will be authenticated and if the user forgets password mail will be generated for user authentication and given option to set new password or users can login through google authentication. All pages check the session cookies to ensure the correct login to the page.

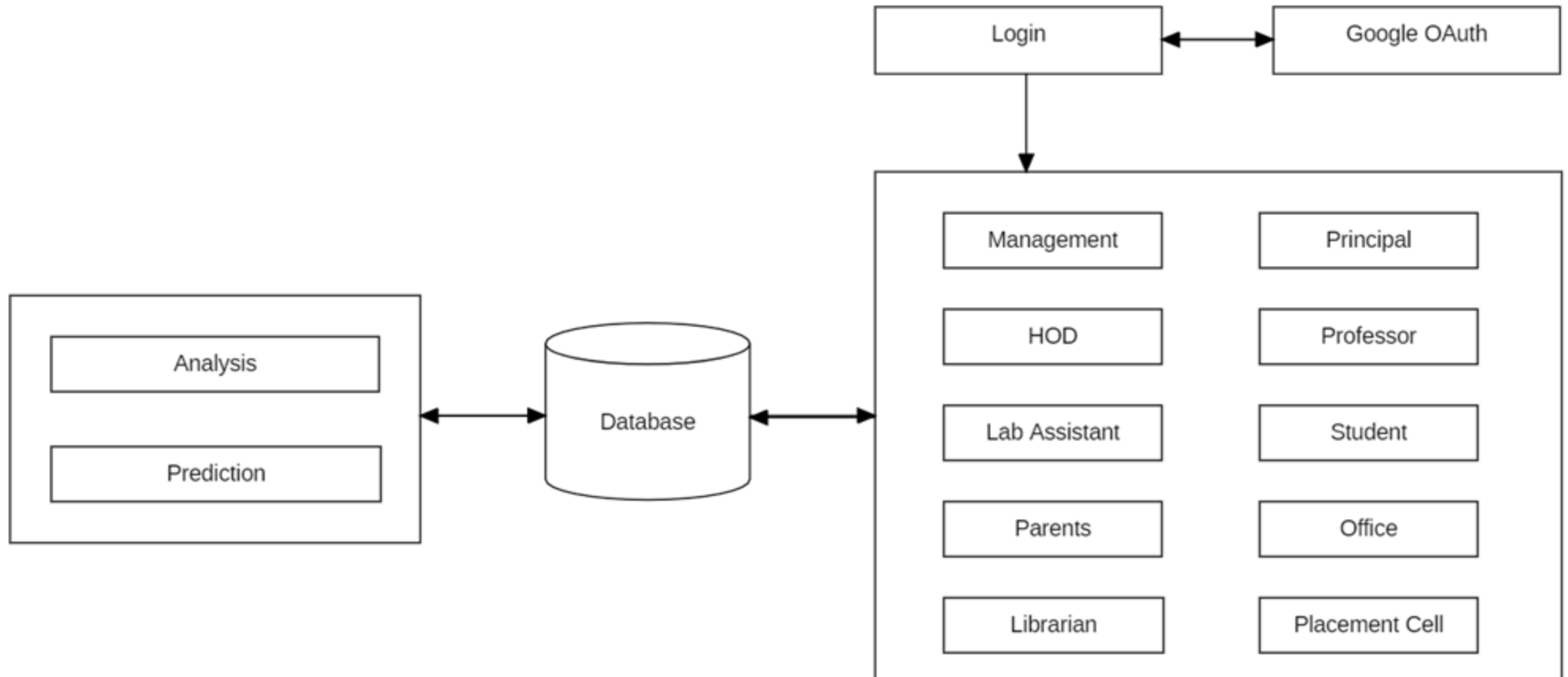
3. Data Management

Since the data is huge in volume, it is stored in different tables particular tables only accessed to improve the speed of retrieval and data integrity is maintained through this.

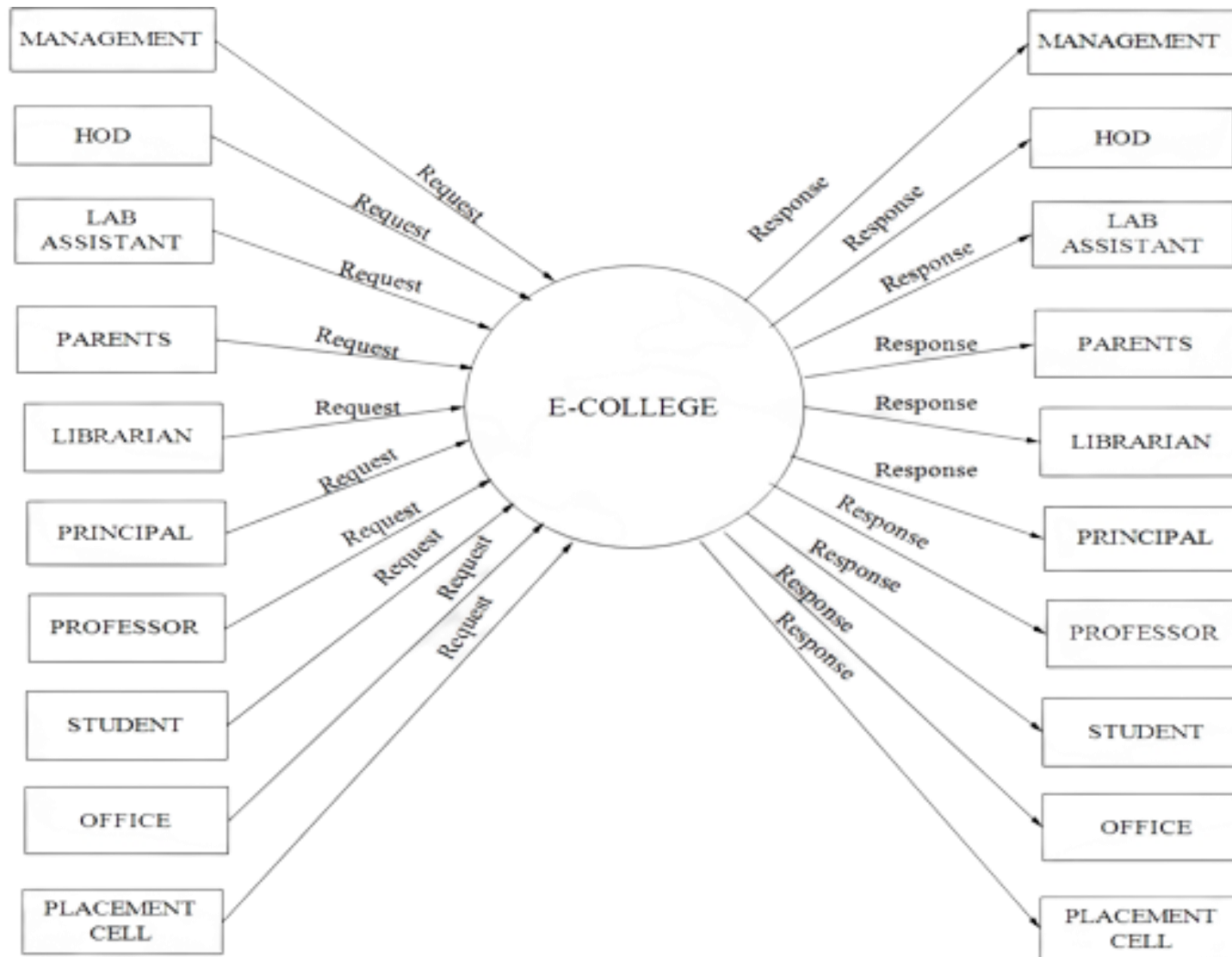
4. Usability

The user interface is simple and minimalistic so users can identify and work without any confusion. The design is responsive so it can fit on any devices without any problem.

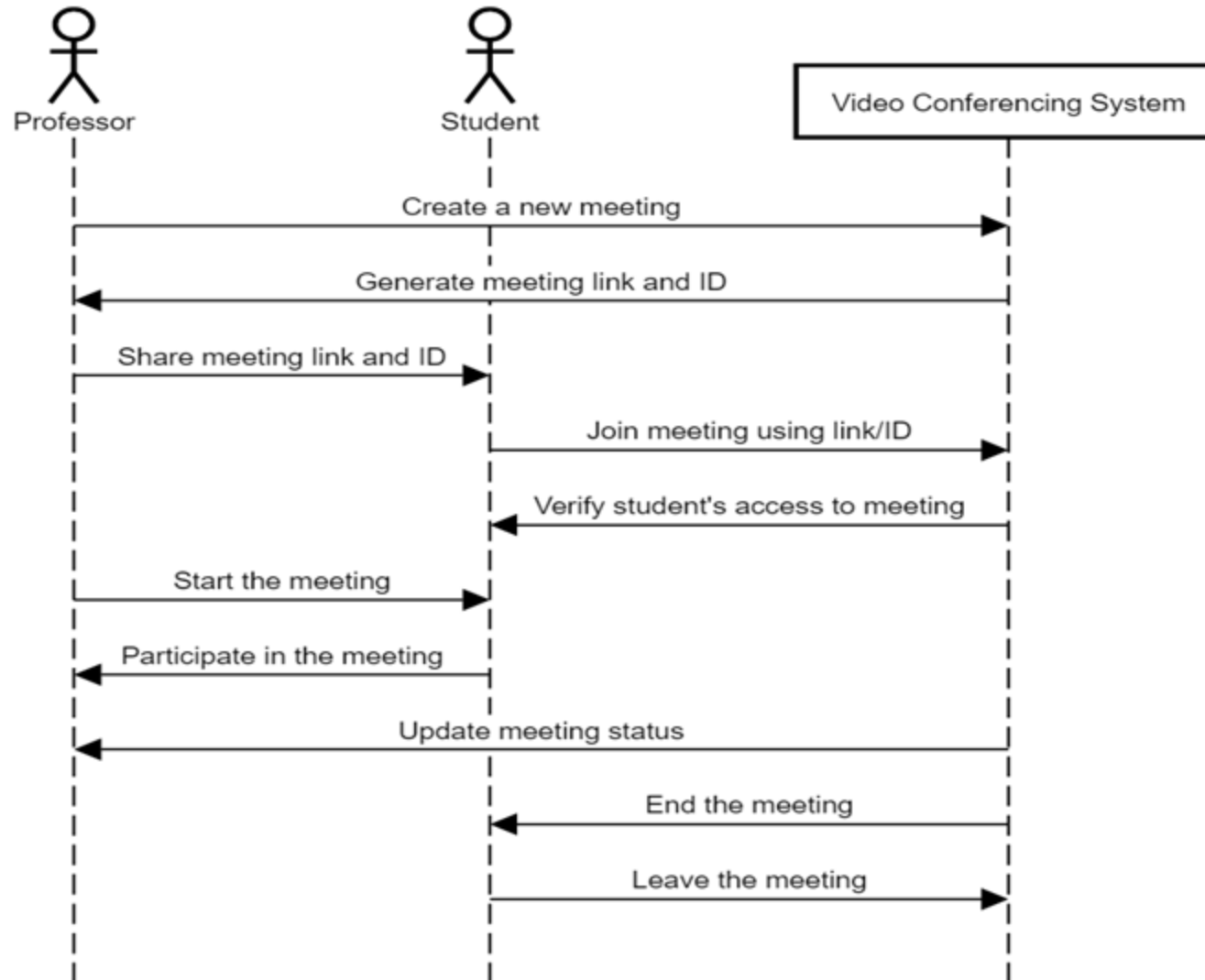
Architecture Diagram



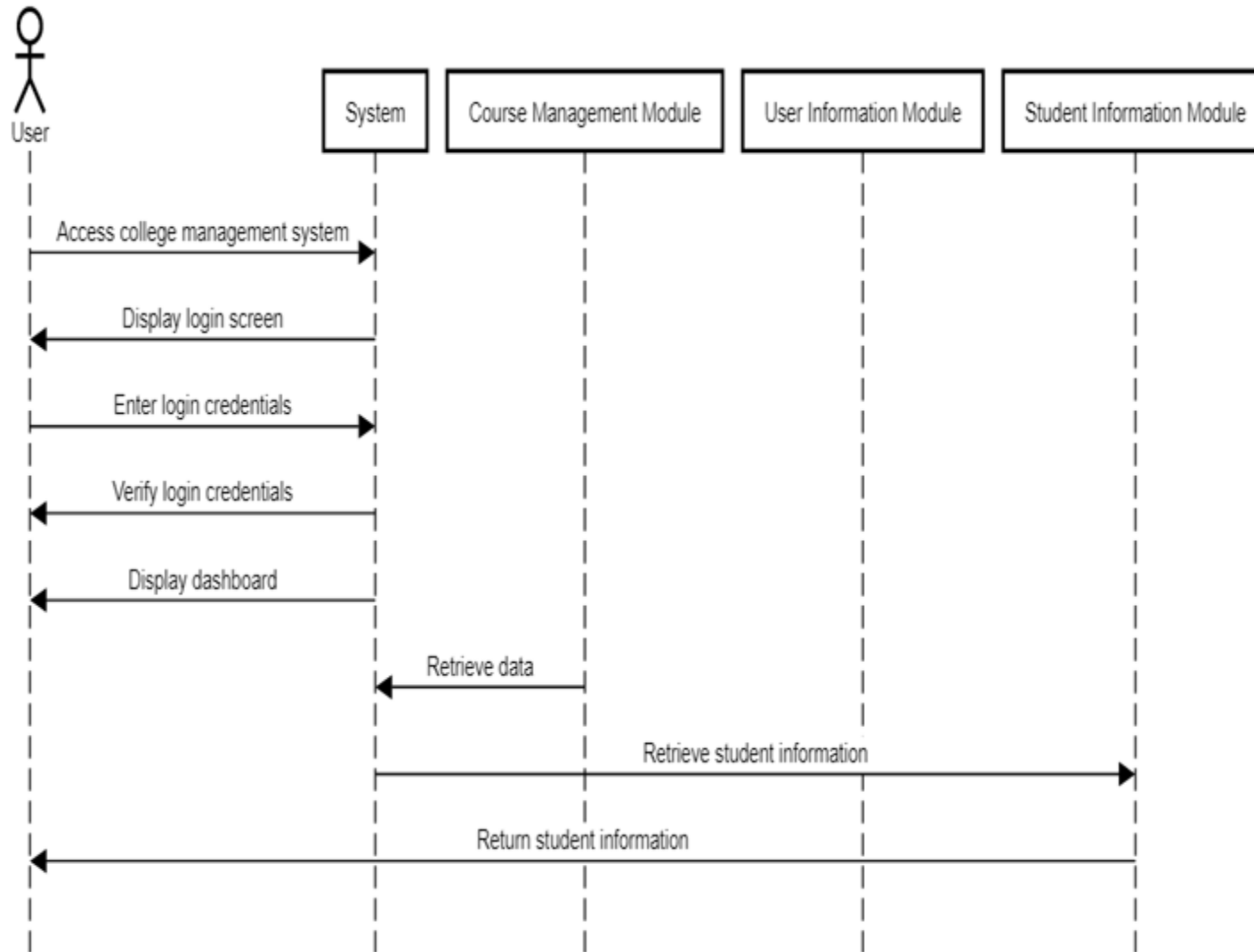
Context Flow Diagram



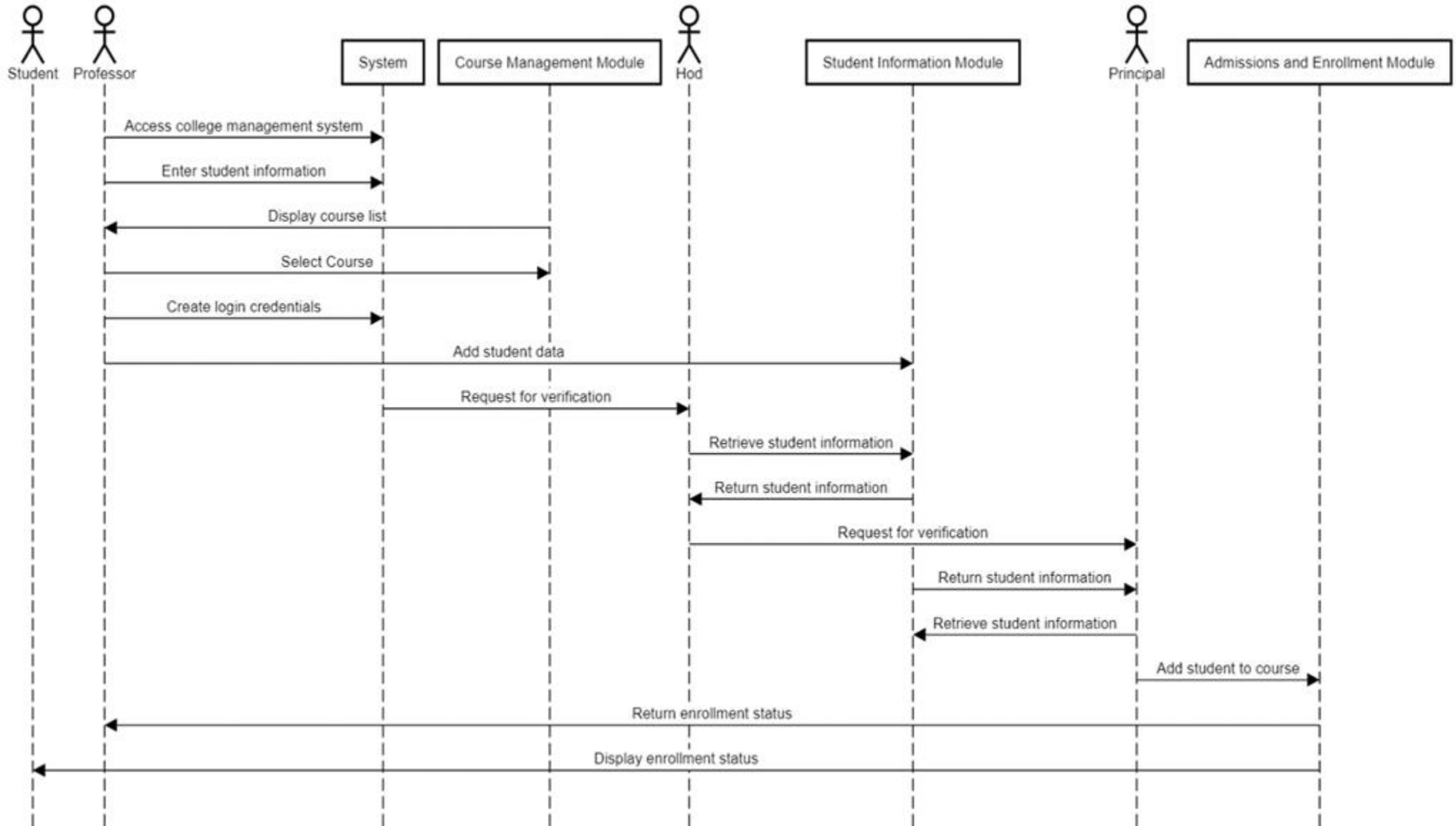
Online meeting sequence diagram



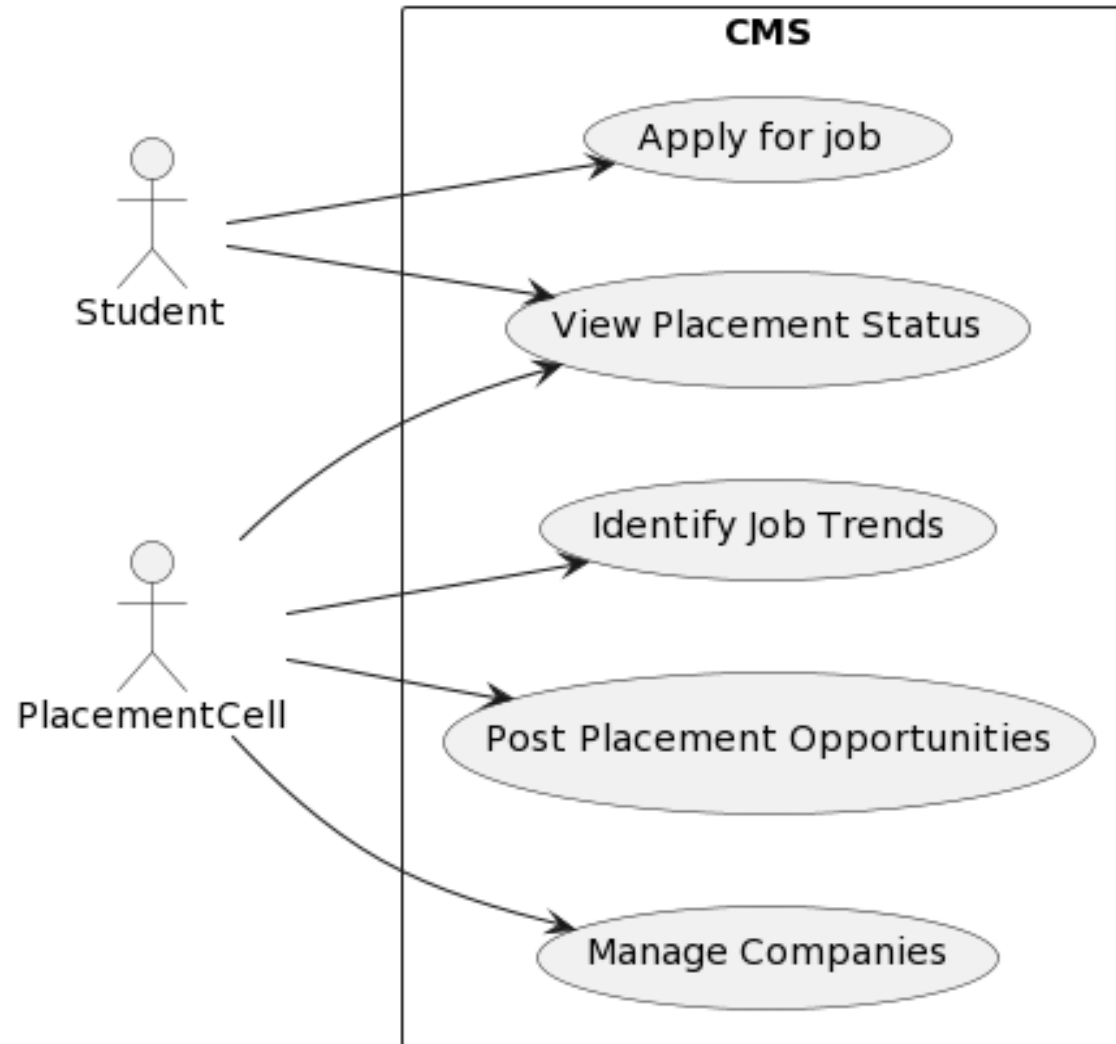
Student Login sequence diagram



Student enrollment sequence diagram



Job portal use case diagram



Implementation

ML Models Used

1. Linear Regression
2. Neural Network

Languages Used

1. HTML
2. CSS
3. JavaScript
4. PHP
5. SQL
6. Node.js
7. Python

Frameworks Used

1. Bootstrap
2. CodeIgniter
3. Express.js
4. Ajax
5. jQuery
6. Streamlit

API Used

1. PyTrends
2. Patym Web Kit
3. AmCharts

Host and Domain Providers Used

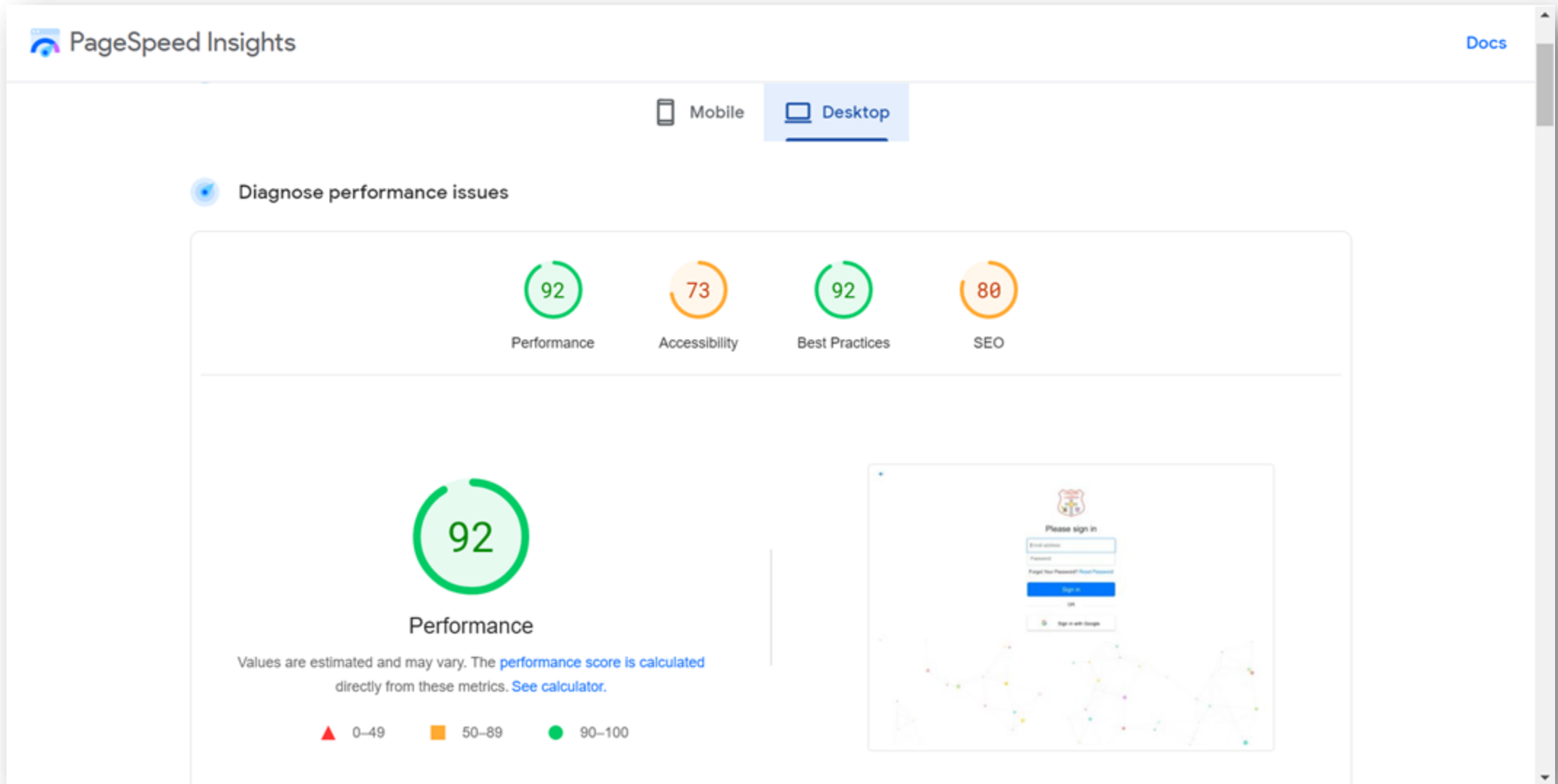
1. Heroku
2. ZegoCloud
3. Infinityfree
4. Freenom
5. Postale.io

Tools Used

1. Visual Studio Code
2. Xampp
3. Google Chrome
4. Pycharm
5. Filezilla
6. Git
7. Git-ftp
8. Heroku CLI

Software Testing

Light House Performance Test



SQL Injection/XSS Test

```
Applications  Places  Terminal  Mar 9 5:49 PM
root@dark-soul: /home/dark-soul

root@dark-soul: /home/dark-soul

[!] legal disclaimer: Usage of sqlmap for attacking targets without prior mutual consent is illegal. It is the end user's responsibility to obey all applicable local, state and federal laws.
Developers assume no liability and are not responsible for any misuse or damage caused by this program

[*] starting @ 17:46:12 /2023-03-09/

[17:46:13] [WARNING] you've provided target URL without any GET parameters (e.g. 'http://www.site.com/article.php?id=1') and without providing any POST parameters through option '--data'
do you want to try URI injections in the target URL itself? [Y/n/q] y
[17:46:34] [INFO] testing connection to the target URL
[17:46:40] [INFO] checking if the target is protected by some kind of WAF/IPS
[17:46:41] [INFO] testing if the target URL content is stable
[17:46:42] [INFO] target URL content is stable
[17:46:42] [INFO] testing if URI parameter '#1*' is dynamic
[17:46:42] [WARNING] URI parameter '#1*' does not appear to be dynamic
[17:46:43] [WARNING] heuristic (basic) test shows that URI parameter '#1*' might not be injectable
[17:46:44] [INFO] testing for SQL injection on URI parameter '#1*'
[17:46:44] [INFO] testing 'AND boolean-based blind - WHERE or HAVING clause'
[17:46:45] [WARNING] reflective value(s) found and filtering out
[17:46:52] [INFO] testing 'Boolean-based blind - Parameter replace (original value)'
[17:46:53] [INFO] testing 'MySQL >= 5.1 AND error-based - WHERE, HAVING, ORDER BY or GROUP BY clause (EXTRACTVALUE)'
[17:46:56] [INFO] testing 'PostgreSQL AND error-based - WHERE or HAVING clause'
[17:47:00] [INFO] testing 'Microsoft SQL Server/Sybase AND error-based - WHERE or HAVING clause (IN)'
[17:47:03] [INFO] testing 'Oracle AND error-based - WHERE or HAVING clause (XMLType)'
[17:47:07] [INFO] testing 'Generic inline queries'
[17:47:07] [INFO] testing 'PostgreSQL > 8.1 stacked queries (comment)'
[17:47:10] [INFO] testing 'Microsoft SQL Server/Sybase stacked queries (comment)'
[17:47:19] [INFO] testing 'Oracle stacked queries (DBMS_PIPE.RECEIVE_MESSAGE - comment)'
[17:47:22] [INFO] testing 'MySQL >= 5.0.12 AND time-based blind (query SLEEP)'
[17:47:26] [INFO] testing 'PostgreSQL > 8.1 AND time-based blind'
[17:47:32] [INFO] testing 'Microsoft SQL Server/Sybase time-based blind (IF)'
[17:47:39] [INFO] URI parameter '#1*' appears to be 'Microsoft SQL Server/Sybase time-based blind (IF)' injectable
it looks like the back-end DBMS is 'Microsoft SQL Server/Sybase'. Do you want to skip test payloads specific for other DBMSes? [Y/n] y
for the remaining tests, do you want to include all tests for 'Microsoft SQL Server/Sybase' extending provided level (1) and risk (1) values? [Y/n] y
[17:48:05] [INFO] testing 'Generic UNION query (NULL) - 1 to 20 columns'
[17:48:05] [INFO] automatically extending ranges for UNION query injection technique tests as there is at least one other (potential) technique found
[17:48:19] [INFO] checking if the injection point on URI parameter '#1*' is a false positive
[17:48:19] [WARNING] false positive or unexploitable injection point detected
[17:48:19] [WARNING] URI parameter '#1*' does not seem to be injectable
[17:48:19] [CRITICAL] all tested parameters do not appear to be injectable. Try to increase values for '--level'/'--risk' options if you wish to perform more tests. If you suspect that there
is some kind of protection mechanism involved (e.g. WAF) maybe you could try to use option '--tamper' (e.g. '--tamper=space2comment') and/or switch '--random-agent'
[17:48:19] [WARNING] your sqlmap version is outdated

[*] ending @ 17:48:19 /2023-03-09/
```

Test Conclusion

Our test results showed that the system is highly efficient and user-friendly. The system's intuitive interface and streamlined workflows allowed users to complete tasks quickly and easily. The system's predictive analytics feature was also effective, providing accurate predictions on student performance and trends in course popularity.

Furthermore, the system's security measures, such as user authentication and data encryption, were effective in protecting sensitive data. Our tests also revealed that the system was highly scalable and able to handle large amounts of data and multiple users without any performance issues.

Screen Shots

Fig 1. Landing Page

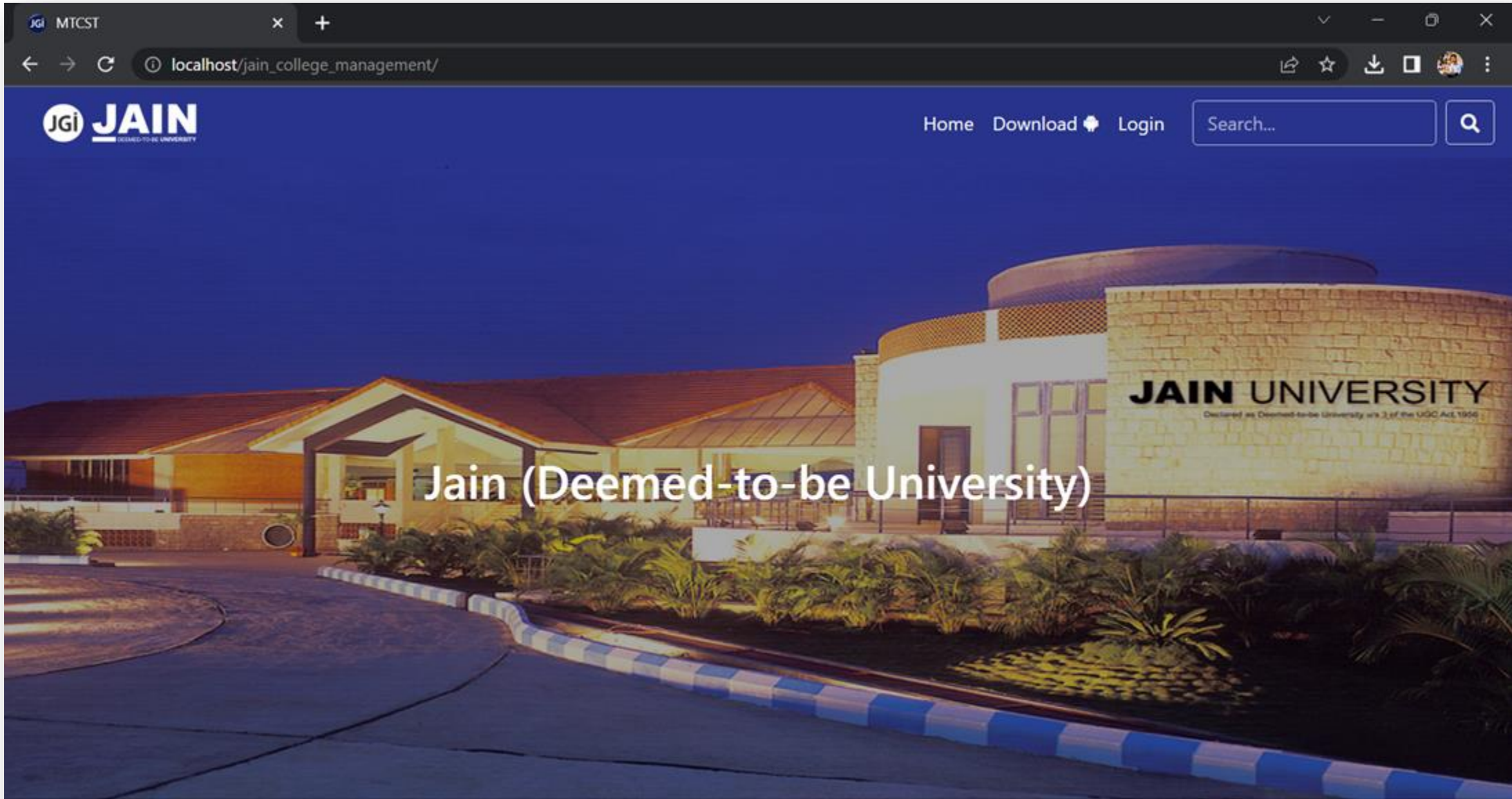
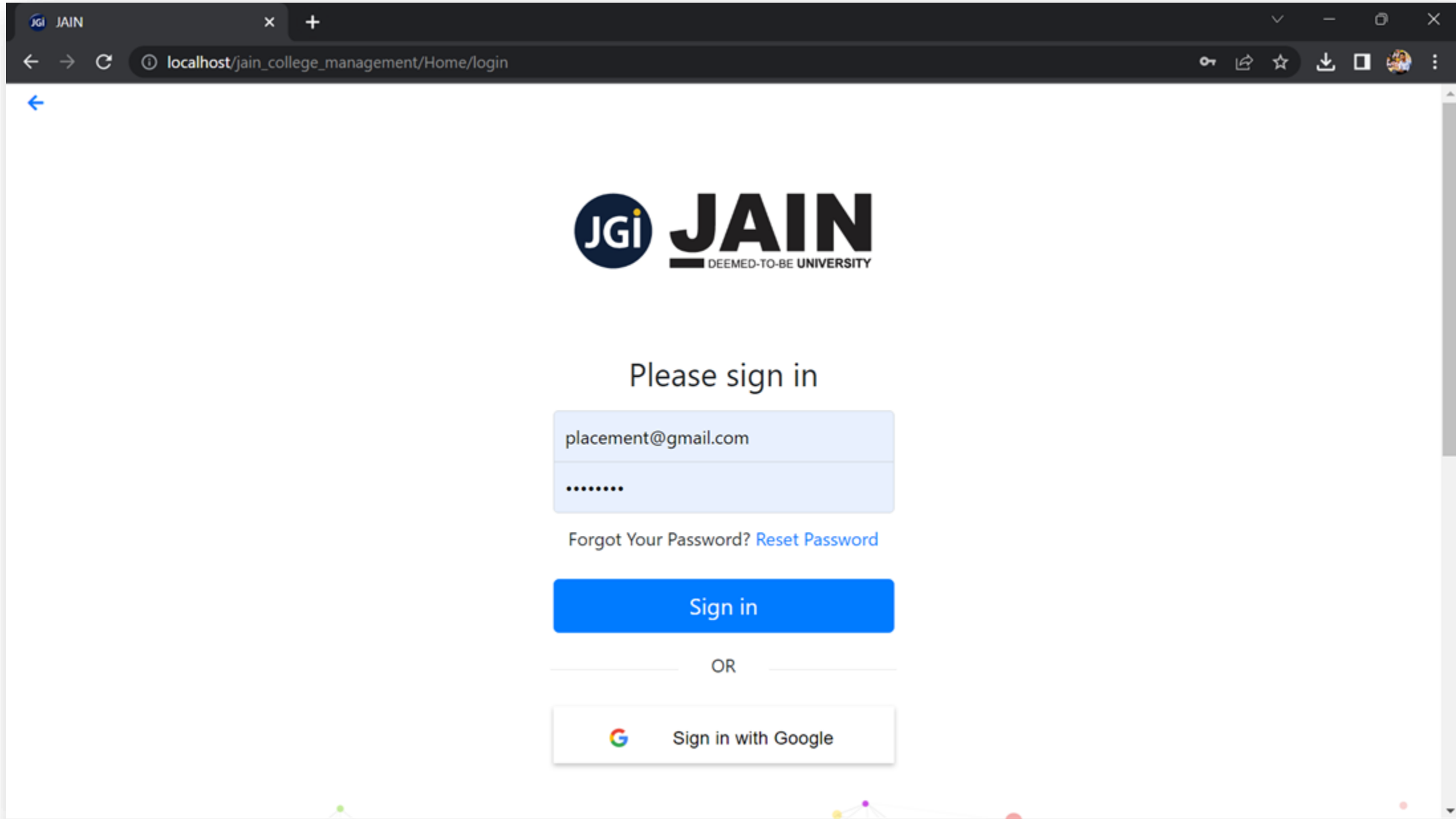


Fig 2. Login Page



The image shows a web browser window with the address bar displaying `localhost/jain_college_management/Home/login`. The page features the JGI JAIN logo, which includes the text "JGI JAIN" and "DEEMED-TO-BE UNIVERSITY". Below the logo, the text "Please sign in" is displayed. There are two input fields: the first contains the email address "placement@gmail.com", and the second contains a masked password ".....". A link "Forgot Your Password? Reset Password" is located below the password field. A blue "Sign in" button is positioned below the "Reset Password" link. Below the button, the word "OR" is centered. At the bottom, there is a "Sign in with Google" button featuring the Google logo.

JGI JAIN
DEEMED-TO-BE UNIVERSITY

Please sign in

placement@gmail.com

.....

Forgot Your Password? [Reset Password](#)

Sign in

OR

 Sign in with Google

Fig 3. Librarian Dashboard

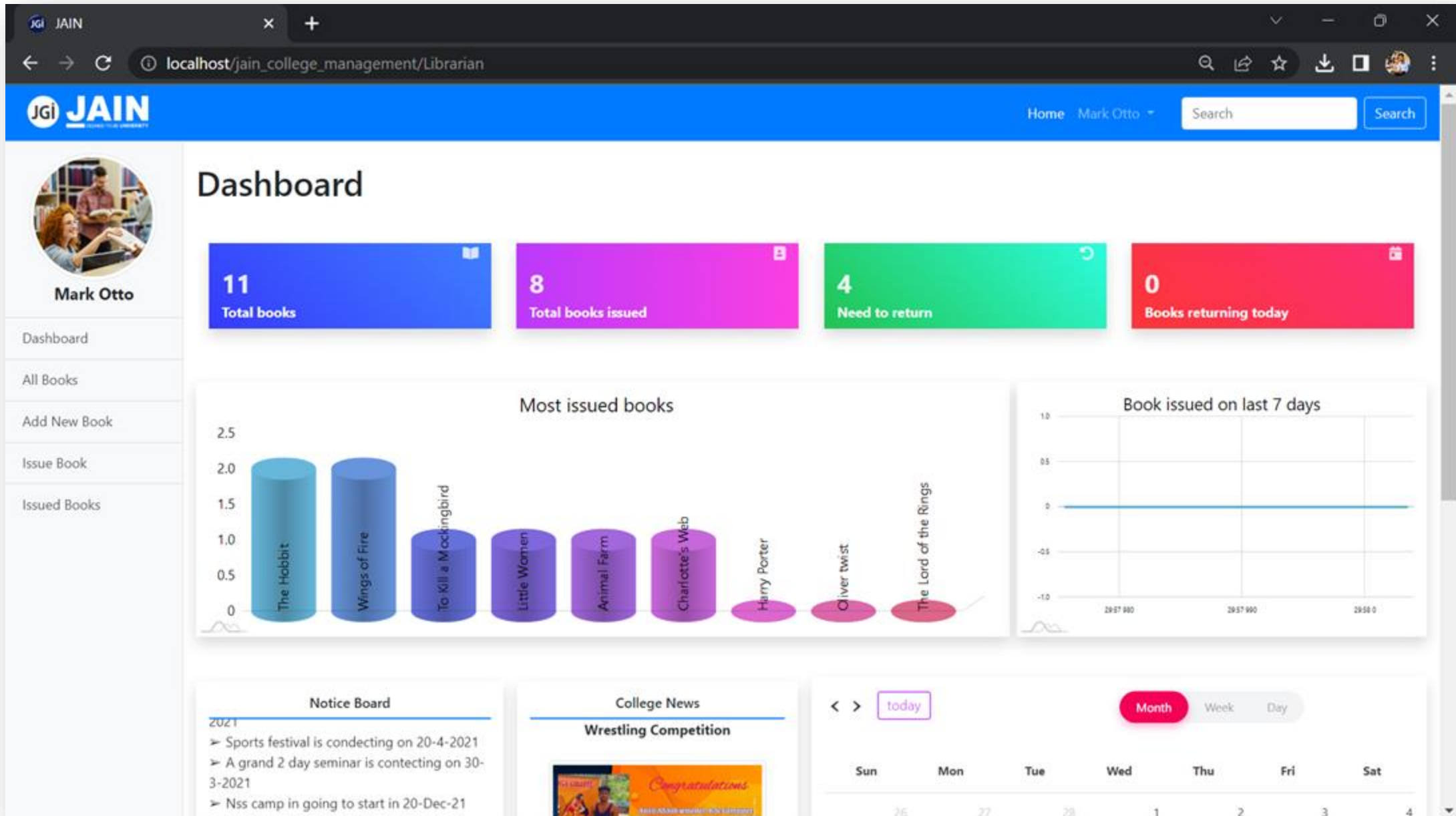


Fig 4. Meeting Page

The screenshot displays a web application interface for JGI JAIN. The browser's address bar shows the URL `localhost/jain_college_management/Professor/live_meeting`. The page features a blue header with the JGI JAIN logo, navigation links for Home and Rachana Jayan, and a search bar. A left sidebar lists menu items: Dashboard, My Students, Add Users, Attendance, Exams, Live Meeting (highlighted with a camera icon), Lab Data, and Assignments. The main content area includes a profile picture of Rachana Jayan, a large video feed of a man, and a form with dropdown menus for Course, Subject, and Semester, along with buttons for Start PTA Meeting and Start Meeting.

JGI JAIN Home Rachana Jayan Search Search

Rachana Jayan

- Dashboard
- My Students
- Add Users
- Attendance
- Exams
- Live Meeting
- Lab Data
- Assignments

Start PTA Meeting

Course Choose... Subject Choose...

Semester Choose... **Start Meeting**

Fig 5. Payments Page

JGI JAIN

localhost/jain_college_management/Student/fee_payment

Home Devadathan R Search Search

No Image Available

Devadathan R

Dashboard

Notes

Exams

Attendance

Assignment

Results

Internal Mark

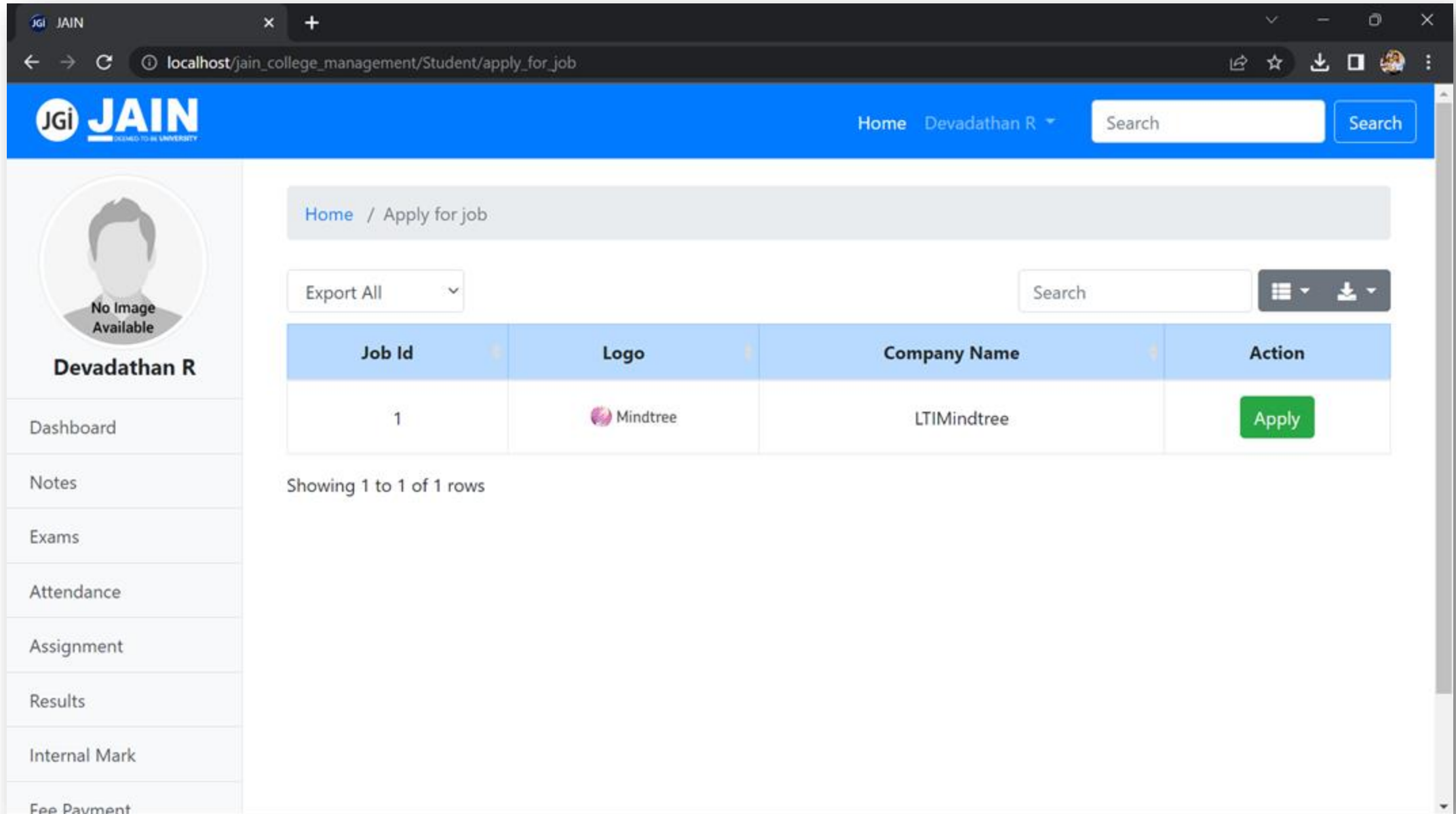
Fee Payment

Home / Fee Payment



#	Semester	Amount	Action
1	Sem 1	0	Pay
2	Sem 2	0	Pay
3	Sem 3	1	Pay
4	Sem 4	1	Pay
5	Sem 5	1	Pay

Fig 6. Job Applying Page



The screenshot shows a web application interface for JGi JAIN. The browser address bar displays the URL `localhost/jain_college_management/Student/apply_for_job`. The page features a blue header with the JGi JAIN logo, a user profile dropdown for 'Devadathan R', and a search bar. A left sidebar contains navigation links: Dashboard, Notes, Exams, Attendance, Assignment, Results, Internal Mark, and Fee Payment. The main content area has a breadcrumb 'Home / Apply for job', an 'Export All' dropdown, a search bar, and a table of job listings. The table has columns for Job Id, Logo, Company Name, and Action. One job is listed with Job Id 1, Mindtree logo, and company name LTIMindtree, with an 'Apply' button in the Action column. Below the table, it says 'Showing 1 to 1 of 1 rows'.

JGi JAIN Home Devadathan R Search

Home / Apply for job

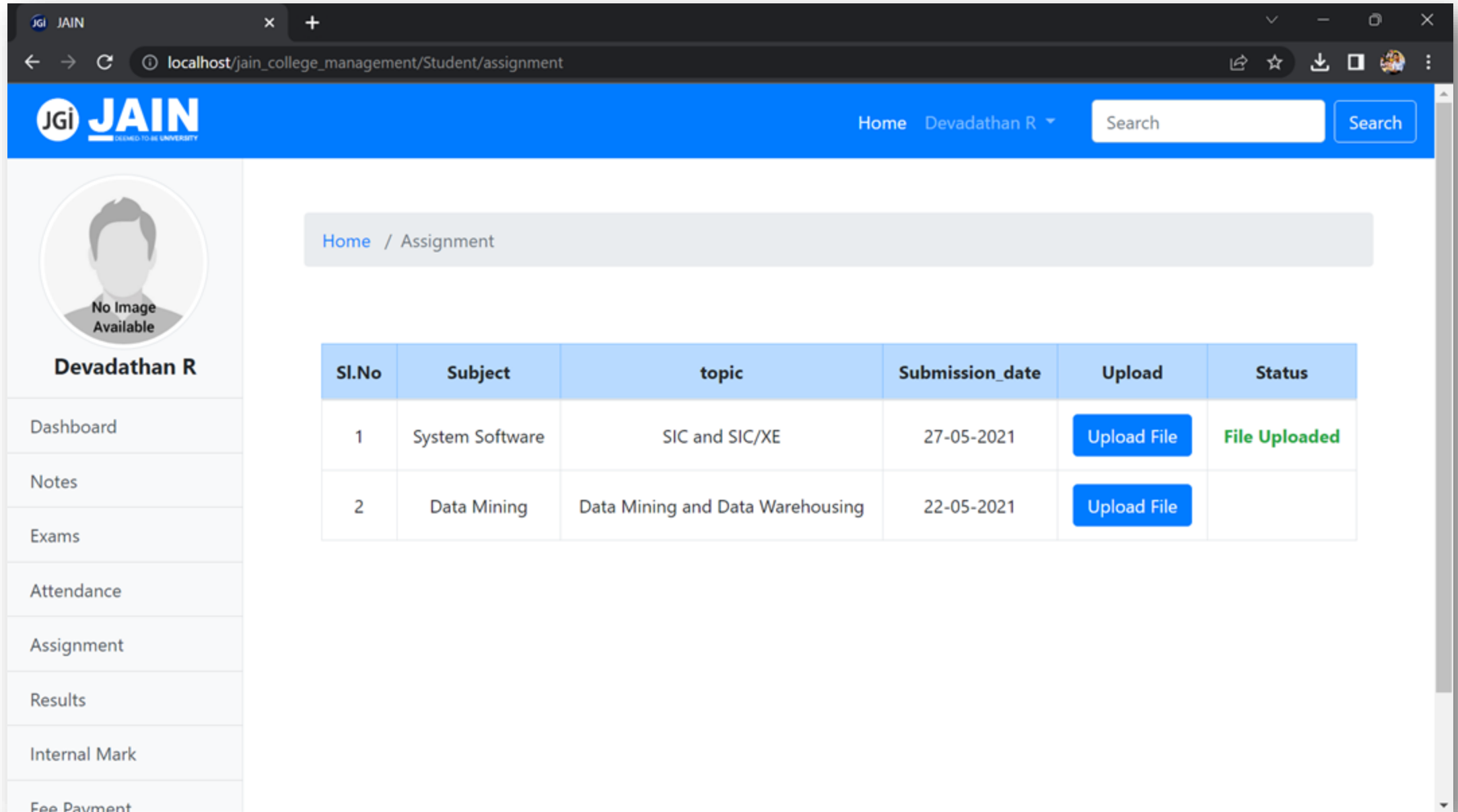
Export All Search

Job Id	Logo	Company Name	Action
1	 Mindtree	LTIMindtree	<button>Apply</button>

Showing 1 to 1 of 1 rows

Dashboard
Notes
Exams
Attendance
Assignment
Results
Internal Mark
Fee Payment

Fig 7. Assignment Uploading Page



The screenshot shows a web browser window with the URL `localhost/jain_college_management/Student/assignment`. The page features a blue header with the JGI Jain logo, a user profile for Devadathan R, and a search bar. A left sidebar contains navigation links: Dashboard, Notes, Exams, Attendance, Assignment, Results, Internal Mark, and Fee Payment. The main content area displays a breadcrumb trail 'Home / Assignment' and a table with assignment details.

Sl.No	Subject	topic	Submission_date	Upload	Status
1	System Software	SIC and SIC/XE	27-05-2021	Upload File	File Uploaded
2	Data Mining	Data Mining and Data Warehousing	22-05-2021	Upload File	

Fig 8. Download Notes Page

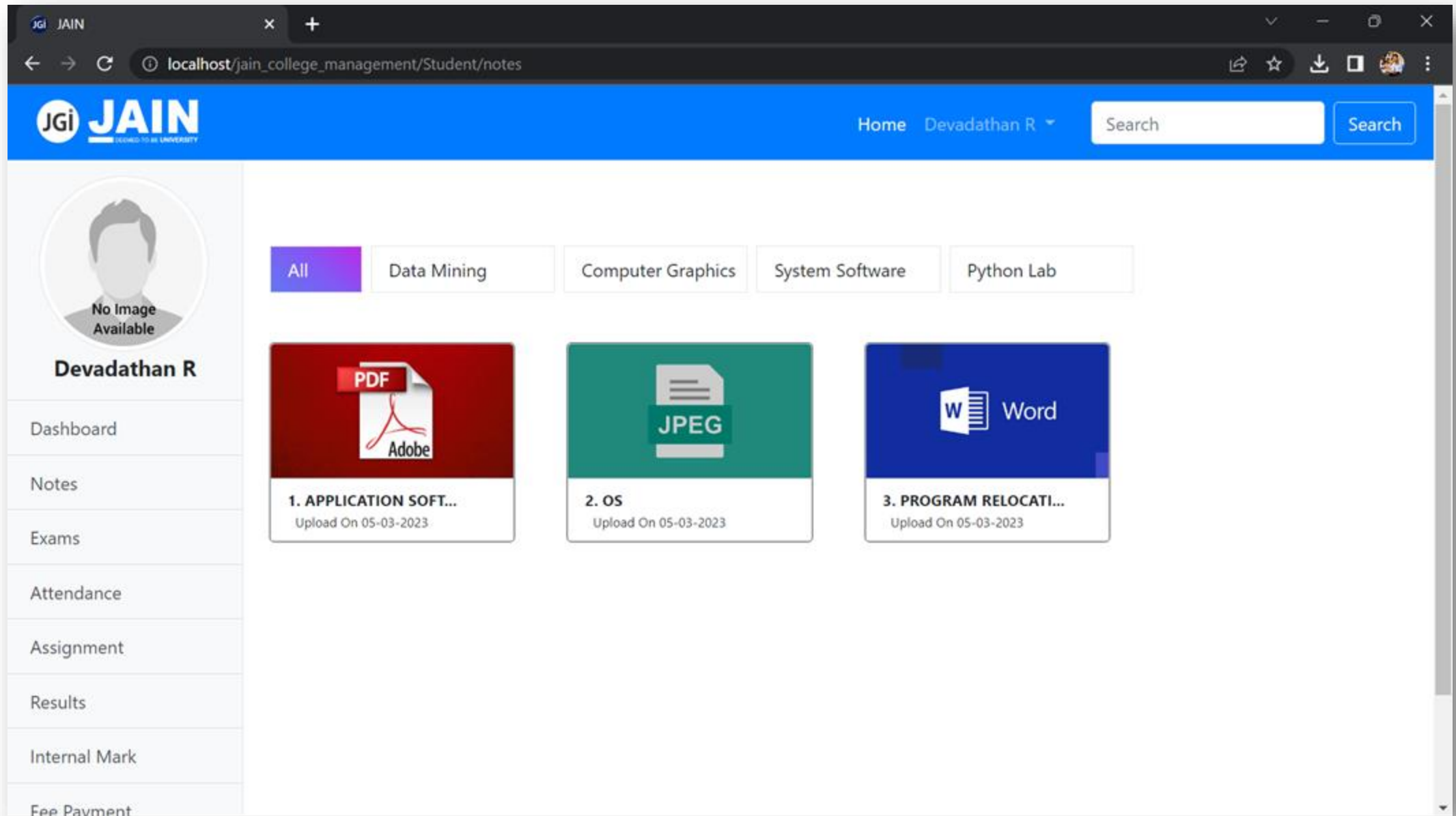


Fig 9. Course Trend Analysis Page

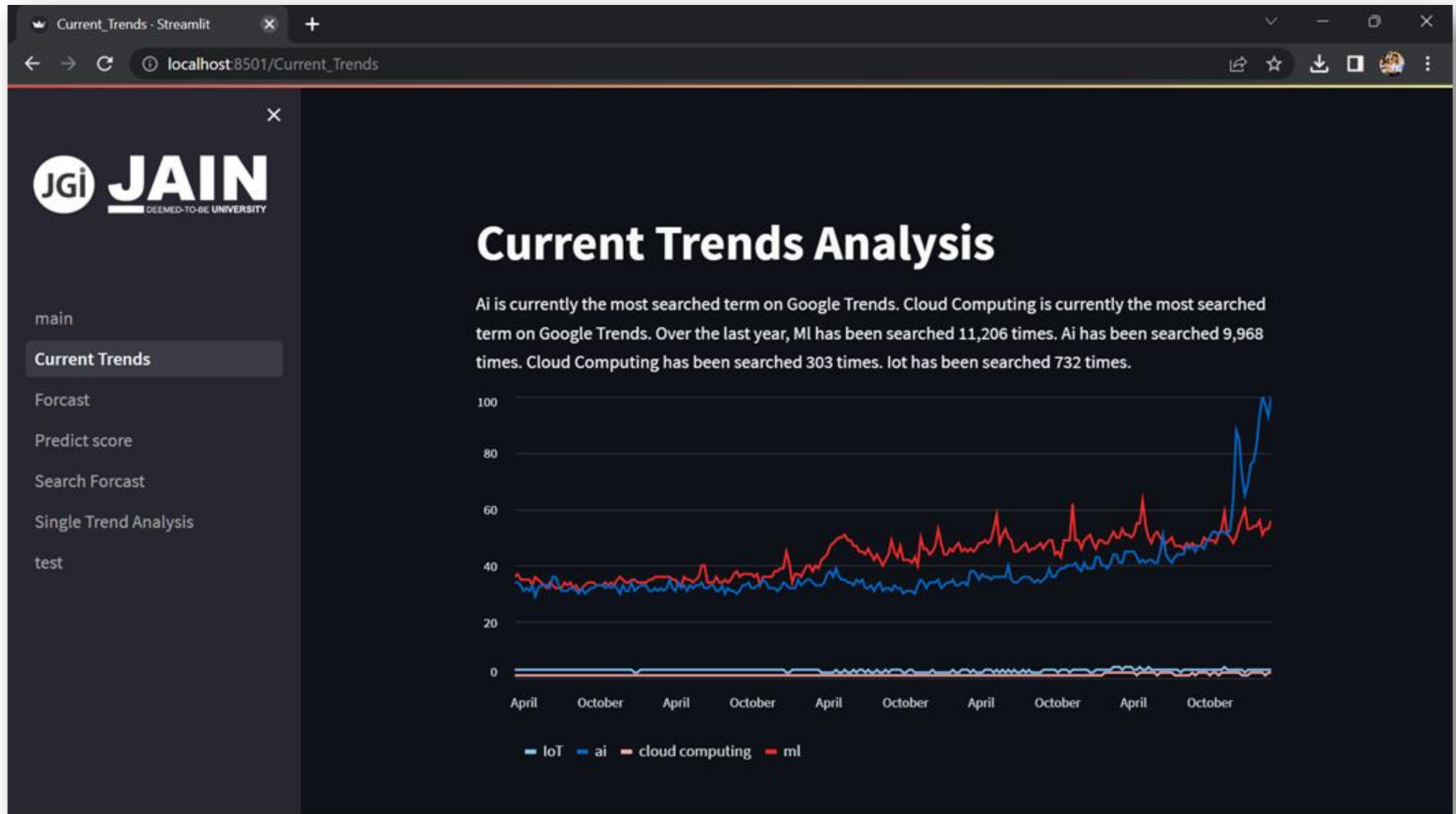


Fig 10. Forecasting Page



Fig 11. Student's Score Prediction Page

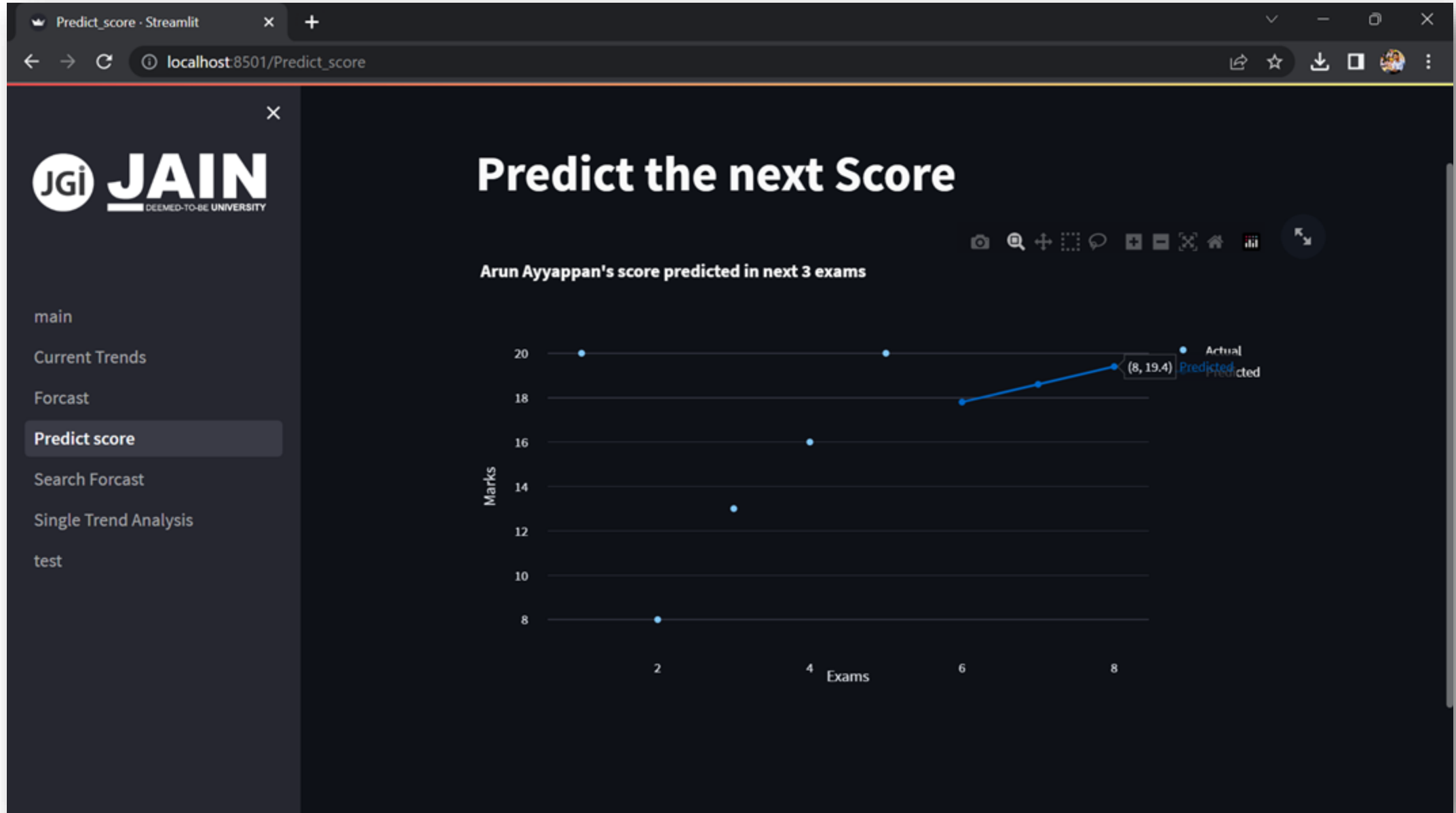
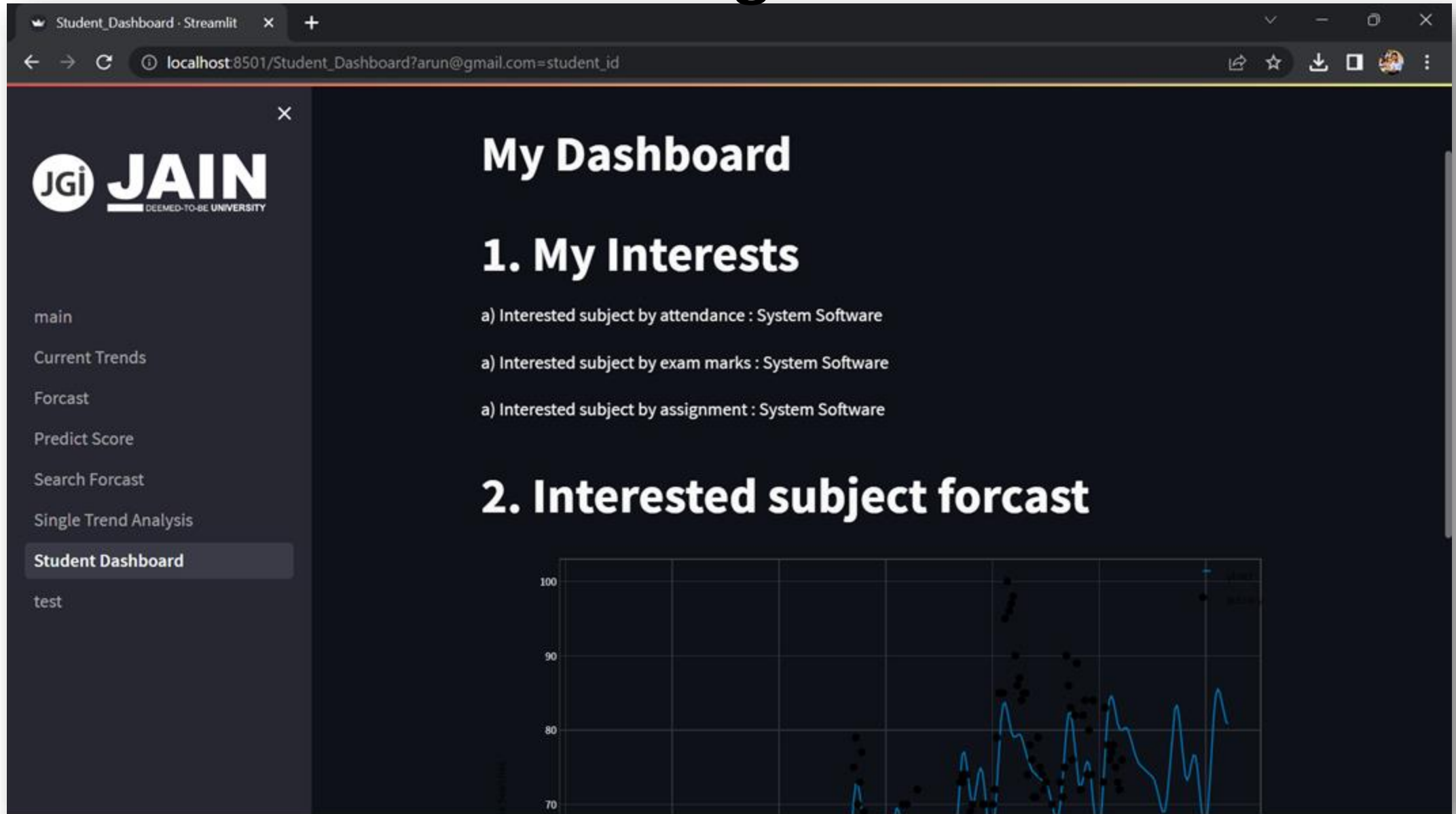


Fig 12. Student s Interested area Forecast Page



Future Enhancement

•There are several potential future enhancements and scopes for this project. Some of them are:

1. Integration with other data sources: The project can be enhanced by integrating data from other sources such as social media, news articles, and other online platforms to get a more comprehensive understanding of the trends and factors affecting the search queries.
2. Multiple models: The project can be further improved by implementing multiple forecast models, such as ARIMA, LSTM, and Prophet, and comparing their performance to choose the best one for forecasting.
3. Real-time forecasting: The current implementation provides a 12-month forecast, but it can be enhanced to provide real-time forecasting with frequent updates. This can be achieved by automating the data extraction and model training process and deploying the model on cloud platforms.

Future Enhancement

4. Visualization and reporting: The project can be enhanced by developing a visualization and reporting module that provides insights into the trends, patterns, and forecasts in a user-friendly manner. This can help users to quickly identify the key insights and make informed decisions.
5. Integration with other applications: The project can be integrated with other applications such as marketing tools, content management systems, and analytics platforms to provide valuable insights and help businesses to make data-driven decisions.

Overall, this project has a lot of potential for future enhancements and scopes, and it can be further improved to provide more accurate and valuable insights for businesses and organizations.

Conclusion

- The new system has overcome most of the limitations of the existing system and works according to the design specification given.
- It consumes less processing power and all the details are updated and processed immediately.
- User Friendly.
- Reduces paper works.
- Combines all the functionalities of a college under a single platform.

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Thank You